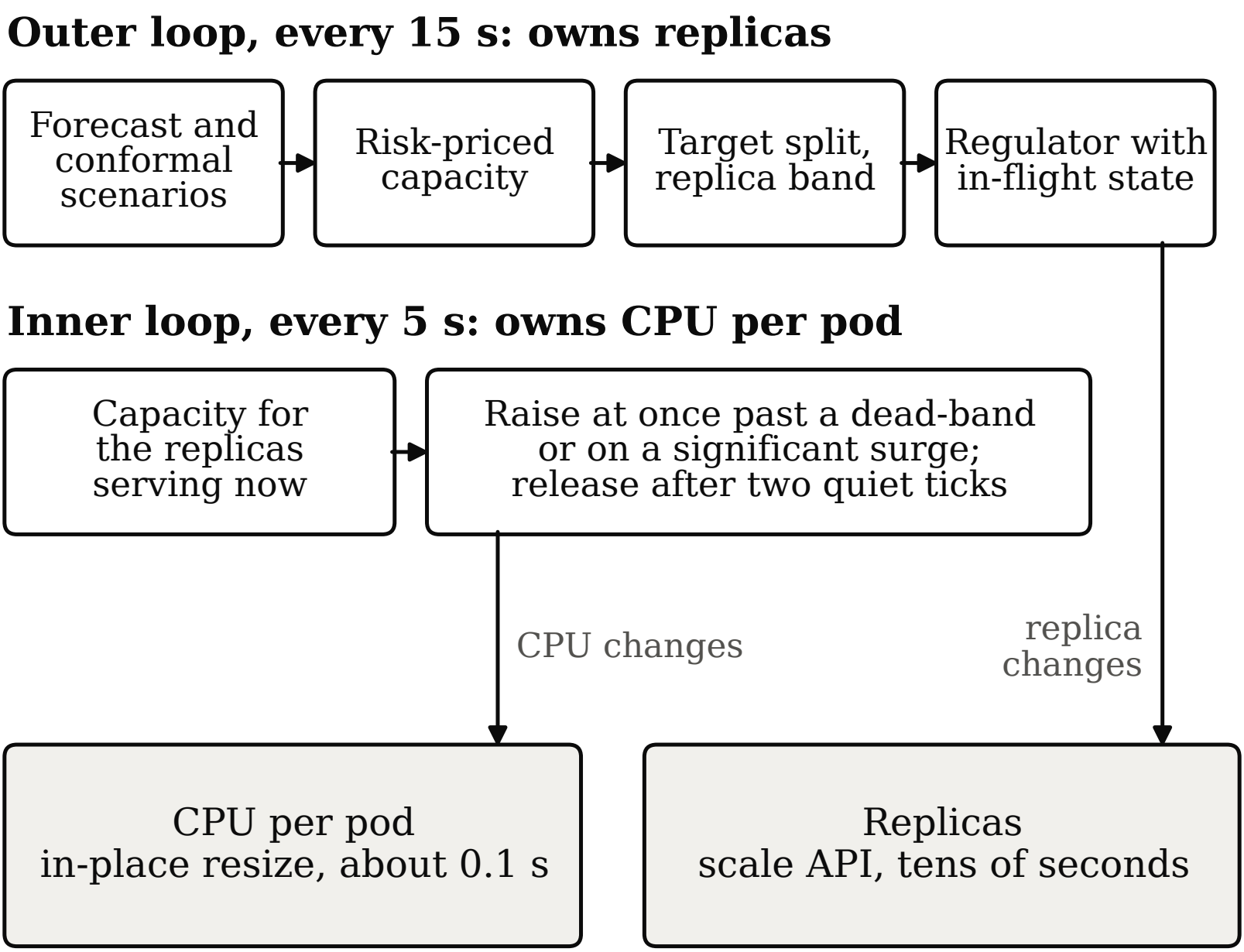
Outer loop, every 15 s: owns replicas
Forecast and conformal scenarios
Risk-priced capacity
Target split, replica band
Regulator with in-flight state
Inner loop, every 5 s: owns CPU per pod
Capacity for the replicas serving now
Raise at once past a dead-band or on a significant surge; release after two quiet ticks
CPU changes
replica changes
CPU per pod
in-place resize, about 0.1 s
Replicas
scale API, tens of seconds